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Convergence in economic growth and institutional quality: Does convergence of institutions matter to catch-up rich economies?

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ABSTRACT

Convergence of economic growth among the global north and south countries has attracted a considerable attention of policymakers and researchers. However, the growth literature lacks the comprehensive empirical evidence on the convergence of institutional quality and its implications for growth convergence particularly in terms of countries' heterogeneities. This study aims to empirically investigate three types of convergence hypotheses both in economic growth and institutional quality. Moreover, it is examined whether the convergence of institutional quality leads to growth convergence? The empirical analysis is based on sample of 120 countries for 1984–2015 period. The findings reveal striking evidence of disparities in speed of growth and institutional quality convergence. The speed of growth convergence is highest in East Asian, transition, and advanced economies; however, the speed of institutional quality convergence is lowest in these countries. Developing countries show the reverse pattern with highest institutional convergence and lowest growth convergence. Such decoupling empirical dynamics between growth and institutional quality convergence indicate the path dependence and lock-in patterns of developing countries. This study empirically shows that higher speed of institutional convergence alone is unable to foster the growth catch-up process.

Introduction

Traditional growth models posit that the cross-country income differences due to disparities in the factor accumulation can be explained by the differences in savings, preferences, and other parameters of economies (Acemoglu et al., 2005). These models show that the diminishing marginal returns to capital enable the poorer countries to catch-up the richer ones in terms of economic growth and income. However, empirical literature on the cross-country income differences and growth convergence exhibits inconsistent patterns. These inconsistencies are largely studied in terms of economic determinants of convergence. North and Thomas (1973, p. 2) argued that innovation and accumulation of factors are proximate causes of economic growth. The fundamental factor behind the cross-country income differences is the difference in institutions. North (1991) stated that the institutions being the rules of game[1] develop the framework of the incentive structure of an economy, as that structure evolves in an economy, it frames the dimension of

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[1] In the words of North (1990): "Institutions are the rules of the game in a society or, more formally, are the humanly devised constraints that shape human interaction." The details on the indicators of institutions used in this study are provided in the section 3 and subsection 5.1.

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economic change towards growth, stagnation or decline. To this end, the role and nature of institutional quality convergence mainly in an empirical fashion have received a little attention. Following the rise of China and growth booms in East Asian economies, we observe the end of great divergence among global South and North countries. This phenomenon unveiled the growth divergence among the global south economies (Frankema, 2025). This phenomenon motivates the research questions of this study by empirically investigating the convergence of growth and institutional quality mainly in a large panel data setting. It is also shown that the speed of growth convergence is higher in East Asian and high-income countries whereas the developing countries exhibit higher speed of institutional convergence.

The fundamental model of growth theory (Solow, 1956; Swan, 1956), hereafter referred as Solow-Swan predicts the exogenous factors such as technology as the foundation-stone of economic growth. Ramsey (1928) and Cass (1965) provided endogenous determinants for economic growth. However, new growth theories by Romer (1986) incorporated the role of human capital and R&D to model the question of economic growth. Grossman & Helpman (1991) argue that technological progress from R&D further improves the growth. Neoclassical growth models for closed economies presented by Ramsey (1928), Solow-Swan, and Cass (1965) posit that per capita GDP growth rates are inversely related to the initial points, such as if economies are identical in terms of preferences and technology, then poor economies will grow faster than richer ones due to law of diminishing returns, which implies the catching-up process. Barro et al. (1991) argue that significant positive association exists between growth rate of real per capita GDP and initial human capital which implies that less fertility rate and high ratio of physical investment to GDP ratio is affected by massive level of human capital. Political stability enhances growth rates whereas market distortions deteriorate growth rates. Economies with less capital to labor ratio embodies higher marginal product of capital catch-up than economies having more capital to labor ratio with less marginal product of capital, due to international mobility of capital and technology. The evidence is stable with Barro and Sala-i-Martin (1992), in terms of capital stock in Cass-Koopmans model, only if elasticity of marginal utility is constant. The catching-up hypothesis is inconsistent with cross-country evidence and shows only little correlation among real per capita growth rates with its initial income level.

Institutions perform in a variety of ways to spur the economic growth of an economy. Findings of well-known institutional economists strongly suggest that political institutions enhance growth (Acemoglu et al., 2001; Hall & Jones, 1999). Aidt (2009), Hall & Jones (1999) and Acemoglu et al. (2001) found that corruption reduces growth in economies with good governance and strong political institutions. Acemoglu et al. (2004, 2008) found that the differences in economic development are due to differences in economic and political institutions, which in turn lead to differences in the prosperity across nations. Lastly, differences in per capita income are caused by differences in economic institutions, more than luck, geography or culture (Acemoglu, 2005). Similarly, Keefer & Knack (1997) and Knack (1996) found that the ability of poorer economies to catch-up the income levels of richer ones is largely determined by the quality of institutions. Economies having less initial capital stock bears the ability to grow faster than the economies having relatively higher stock of capital. Countries having initially low per capita income should have some social capabilities to absorb the technology of advanced economies, capital stock and access to global market for the purpose of catching up. Lucas (1990) stated that the puzzle of capital flows inadequacy can only be resolved by assuming identical marginal product of capital across nations and he argued that differences in human capital, external benefits of human capital and imperfect capital markets are major factors hindering capital flows in poorer economies.

Following the above discussion, it is evident that beside the proximate determinants, institutional quality plays a crucial role in the long-run determination of output and growth, however it is still unclear that to what extent the convergence of institutional quality contributes to the growth convergence. Building upon such empirical gaps in the growth convergence literature, this study aims to examine the patterns of and the association between growth and institutional quality convergence. The empirical results from the conditional convergence offer practical insights in terms of narrowing institutional quality disparities coupled with rising growth differences among the developing countries. The convergence literature mainly provides theoretical aspects of vitality of institutional quality for growth and convergence but fails to provide empirical evidence on how does institutional quality convergence matter for growth convergence. Thus, the core objectives of this study are to empirically investigate the growth and institutional convergence. The third research question is addressed simply from the implications of empirical outcomes of first two research questions. This study empirically addresses the following research questions.

- Does there exist growth convergence among the selected countries?
- Does there exist institutional convergence among the selected countries?
- Does institutional convergence lead to growth convergence among the selected
- countries?

Rest of the study is organized as follows; section 2 provides a comprehensive review of related literature. Section 3 presents the data and variables description. Estimation methods and theoretical framework are provided in section 4. Section 5 provides the estimation results and discussions. Finally, the conclusion and policy implications are given in the last section.

Review of literature

How fast do economies converge?

The findings of Barro and Sala-i-Martin (1992), Barro et al. (1991) and Mankiw et al. (1992) reveal that the economies converge to their balanced growth path with an average rate of 2 per cent a year, which is quite low due to only one reproducible factors of

production which is physical capital. They concluded that the human capital must be qualitatively identical to that of physical capital and also the elasticity of output with respect to all reproducible factors must be about 0.8 which is larger than physical capital alone. Although the results obtained are consistent with the neoclassical growth model but are based on some unrealistic assumptions. In contrast, Evans (1997) stated that the results favoring the neoclassical growth model are consistent only in case of restrictive assumptions, and an alternative approach produces consistent estimates under weak assumption. Evans showed that the speed of convergence is enhanced to 5.89 per cent a year for the sample of 48 countries, and 15.50 per cent a year for US states. Alesina et al. (1996) found strong positive relationship among economic growth, capital accumulation political freedom and democratic institutions and negative among economic growth, political instability, and income inequality. It is still unclear that what if some countries exhibit higher institutional quality convergence but experience relatively slow convergence in growth. We empirically focus on this aspect of the convergence and speed of convergence in growth and institutional quality.

Barro (1996) found that higher level of initial schooling and life expectancy, lower inflation, less government consumption, political freedom, improved terms of trade and rule of law, and democracy has strong positive influence on the standard of living on a country's prosperity, implies long-run determination to overall economic growth. Beyaert (2003) found that there exists strong evidence of convergence of real per capita GDP due to their mean reverting property of per capita output. The augmented Solow model by Mankiw et al. (1992) exhibited the role of human capital for long term economic growth assuming identical technology across economies and taking technological differences as residuals for growth. However, Hall and (Jones, 1996, 1999) and Skidmore and Deller (2008) found that the competition at regional level leads to convergence of per capita output for factors of production. The empirical finding shows strong evidence for the convergence of government spending at the local level. Balcerowicz (2007) found that borrowing and adoption of broadly defined technologies, stemmed from the advanced economies, are the proximate cause of convergence and to explain the differences among economies real income levels. Rassekh et al. (2001) concluded that the convergence in the OECD economies after the postwar period occurs through investment, government consumption and export. Beyaert (2003) empirically found no evidence of convergence for the European Union sample. However, the empirical evidence shows only a little bit evidence of conditional convergence, and hence reduces regional inequalities, for short run only. Similarly, Safdar and Nawaz (2020) examined the growth convergence for panel of South Asian countries and found no empirical evidence of growth convergence among these countries. Sandilah and Yasin (2011) empirically found no absolute β-convergence and σ-convergence in Pakistan and also in rural-urban areas, except for conditional convergence. Young et al. (2008) argues that β-convergence is statistically significant across the United States and many of its individual states.

Literature on institutional quality convergence

The ability of poor economy to catch-up the income levels of richer ones is largely determined by the quality of institutions (Keefer and Knack, 1997; Knack, 1996). Institutions are path dependent in a sense that group of economies having high-quality institutions today, would largely be the same in the future. Savoia and Sen (2016) concluded that regardless of the initial conditions, institutional quality has demonstrated slow convergence resulting from major historical events such as, the end of the Cold War. Institutional catch-up is significantly higher amongst the economies of identical structures implying the conditional convergence. Mitra et al. (2008) concluded that the growth patterns at the sectoral and firms level indicate a considerable convergence and strong evidence of convergence exists in the transition economies that joined the European Union. Grusevaja and Pusch (2015) concluded that there exists inertia for the institutional variables which indicates weak evidence of institutional convergence.

Iancu (2009) argued that efficiency and impact of institutional quality and capital are used to fill the economic gap between economies in the context of enforcement and implementation of European Integration strategy. It implies strong evidence of convergence among the EU economies. Karasan (2015) highlighted the importance of productivity and institutions in the convergence process of economic growth. The results reveal that estimated speed of conditional convergence is relative higher in productivity analysis. Keefer and Knack (1998) concluded that poor institutional quality is the root cause of the economic divergence. They have also analyzed the different indicators of institutional quality e.g., rule of law, corruption, risk of expropriation and contract repudiation. It was found that institutional environment in which economic activity takes place, determines in large part the ability of poor countries to catch-up. Knack and Keefer (1995) reported three propositions; first, institutional quality has insufficient proxies of political and civil liberties and political violence, which protects property rights. Second, institutions that protect property rights are crucial to economic growth and investment. Third, strong evidence of conditional convergence emerges after controlling for institutional quality. Knack (1996) concluded that there exists strong convergence in income per capita among the economies with secure property rights.

The growth and convergence literature demonstrates inconclusive evidence on nature and patterns of convergence and lacks empirical findings focusing the convergence of institutional quality and its effects on cross-country growth differences. Thus, one of the core objectives of this study is to provide global empirical evidence on the convergence of economic growth and institutional quality and how does the convergence of institutions matter for the growth convergence.

Data & variables description

Empirical analysis is based on sample of 120 countries for 1984–2015 period. We have divided the sample into 6 sub-groups: advanced, developing, East Asian Tigers, EU, G7, and Transition Economies, following IMF classification of countries. This classification is based on level of real per capita GDP, export diversification and degree of integration into the global financial system. Real per capita GDP, investment, saving, government expenditure, trade openness, education, population growth rate, inflation rate, and R&D

expenditure data are taken from the (World Bank, 2016). Data of human capital are taken from Feenstra et al. (2015), while economic freedom and polity 2 (political democracy) are obtained from Freedom House Dataset (House, 2019). Role of military in power & politics, government stability, corruption in government, bureaucratic quality, law & order, religion in politics, ethnic fractionalization/tensions, used International Country Risk Guide (ICRG) dataset (PRC Group, and others, 2025). Governance indicators (voice & accountability, political stability & absence of violence/terrorism, government effectiveness, regulatory quality, rule of law, control of corruption) are extracted from World Governance Indicators (Kaufmann et al., 2010). Quality of legal system and security of property rights uses Fraser Institute Index of Economic Freedom. Data of legal origin dummies are taken from Andrei Shleifer's website and distance from the equator simply uses internet source. We used growth rate of real per capita GDP to examine the convergence of per capita GDP. Investment, saving, government expenditure, and trade openness are used as ratio of GDP. Moreover, education is proxied by secondary school enrollment rate. R&D expenditure is used as percentage of GDP. Human capital is calculated from years of schooling and return on education. Economic freedom index is comprised of four major pillars namely, rule of law, open market, limited government role, and regulatory efficiency.

Estimation methodology and theoretical background

This study empirically investigates the three types of convergence, namely, sigma convergence, absolute/unconditional beta convergence, and conditional beta convergence. First, sigma convergence indicates that economies are converging in a sense that the dispersion of their real per capita GDP or institutional quality levels is diminishing over time (Mulligan and Sala-i-Martin, 1993). Absolute/unconditional beta convergence reveals that how the poor economies of the world are growing faster in terms of real per capita GDP or institutional quality levels than the rich countries. Conditional beta convergence shows that countries are converging to each other in the long run in terms of both real per capita GDP and institutional quality given the certain country characteristics.

4.1. Growth convergence models

For all types of growth convergence tests, we have followed the models proposed by Chowdhury (2004).

a) **Sigma Convergence**

Sigma growth convergence is calculated by estimating the Eq. (1).

$$\boldsymbol{\sigma_t = \alpha + \beta_t + \mu_t} \tag{1}$$

In Eq. (1), α and β are the parameters, and μ_t is the stochastic error term. A significant negative (positive) value of the parameter β implies convergence (non-convergence). For the σ_t we have used y_{it} as the growth rate of real per capita GDP. i stands for country and t for the time period, thus σ_t would be the standard deviation of y_{it} across i at time t.

a) **Absolute/Unconditional Beta Convergence**

Absolute/unconditional beta convergence is obtained by estimating Eq. (2),

$$\boldsymbol{\left(y_{it} - y_{i,t-T}\right) = \alpha + \beta y_{i,t-T} + \mu_t} \tag{2}$$

In Eq. (2), dependent variable is the annual growth rate of real per capita GDP, α and β are the parameters to be estimated, μ_t is a zero mean error term. A significant negative (positive) value of β implies convergence (non-convergence). I $(i = 1,2,3....N)$ denotes country, t stands for the end time period and $t - T$ is the beginning of period. The magnitude of β indicates the speed of convergence (divergence).

a) **Conditional Beta Convergence**

In general, outcomes of unconditional convergence suggest that differences in real per capita GDP levels between countries are narrowing over time, but this might be rather a slow process due to diverse structural characteristics prevailing in those countries. Would this process be faster among countries that share the same structural characteristics? Answering this question requires testing the conditional convergence hypothesis. Technically, conditional convergence states that poorer countries tend to grow faster than the richer ones if they share common characteristics, such as savings, human capital, innovation, and institutional quality. This hypothesis is tested with Eq. (3).

$$\boldsymbol{\left(y_{it} - y_{i,t-T}\right) = \alpha + \beta y_{i,t-T} + \gamma \chi_i + \mu_t} \tag{3}$$

In Eq. (3), dependent variable is the annual growth rate of real per capita GDP, α and β are the parameters to be estimated, μ_t is a zero mean error term. χ_i is a set of control variables that account for long-run determinants of real per capita GDP change across countries. The control variables used in the conditional convergence regression include initial value of per capita GDP growth rate, human capital, saving, investment, trade openness, R&D, population growth rate, institutional quality, economic freedom, polity2,

military in power and politics. Inclusion of the control variables ensures that if all of these characteristics remain same, what happens with the cross-country income differences? In other words, the Eq. (3) provides the convergence estimates given that all of the above characteristics (control variables) do not change. A significant negative (positive) value of β implies convergence (non-convergence), only when $\gamma \neq 0$. Again, the magnitude of β expresses the speed of convergence (divergence).

Institutional quality convergence models

For institutional quality convergence tests, we have followed the models proposed by Savoia & Sen (2016).

a) **Absolute/Unconditional Beta Convergence**

Absolute/unconditional beta convergence is obtained by estimating Eq. (4), Let ($\boldsymbol{G_{it}}$) denotes the observed institutional quality measure in country i observed at both date $t = 0$ and $t = T$, that is, at the beginning and at the end of the sample period, respectively. A test equation for institutional quality convergence is given below.

$$\frac{(\boldsymbol{lnG_{iT}} - \boldsymbol{lnG_{i0}})}{\boldsymbol{T}} = \boldsymbol{\alpha} + \boldsymbol{\beta G_{i0}} + \boldsymbol{\varepsilon_i} \tag{4}$$

In the Eq. (4), the dependent variable is the average annual growth rate of institutional quality, α and β are the parameters to be estimated, $\boldsymbol{\varepsilon_i}$ is a zero mean error term. A significant negative (positive) value of β implies convergence (non-convergence). Moreover, the magnitude of β shows the speed of convergence.

a) **Conditional Beta Convergence**

Similar to the conditional Beta convergence model of growth, the conditional convergence in the institutional quality show that the differences in institutional quality among countries decrease over time if they share common characteristics, such as real per capita GDP growth rate, polity2 (political democracy), human capital, investment, R&D, religion in politics, ethnic fractionalization, geography (log of distance from the equator in KM), and legal origin dummies. This hypothesis is tested based on Eq. (5).

$$\frac{(\boldsymbol{lnG_{iT}} - \boldsymbol{lnG_{i0}})}{\boldsymbol{T}} = \boldsymbol{\alpha} + \boldsymbol{\beta G_{i0}} + \gamma\chi_{i0} + \boldsymbol{\varepsilon_i} \tag{5}$$

In the Eq. (5), dependent variable is the average annual growth rate of institutional quality, α and β are the parameters to be estimated, $\boldsymbol{\varepsilon_i}$ is a zero mean error term. χ_{i0} is a set of control variables that account for long run determinants of institutional quality. The vector of control variables include initial values of institutional quality measure (governance index), real per capita GDP growth rate, polity2 (political democracy), human capital, investment, R&D, religion in politics, ethnic fractionalization, geography (log of distance from the equator in KM), and legal origin dummies. These control variables allow us to estimate the convergence conditional on these country characteristics. Note that the above variables also include the instrumental variables used in the 2SLS based conditional convergence models. A significant negative (positive) value of β implies convergence (non-convergence), only when $\gamma \neq 0$. The magnitude of β expresses the speed of convergence (divergence).

Results and discussion

Descriptive analysis

5.1.1. Institutional Quality Convergence

Since the institutional quality is a persistent phenomenon, we examined trends of legal, bureaucratic, and administrative institutional quality measures over the long periods, by focusing on cross-country dataset. Institutional quality used in our study follows the average of four indices, namely, quality of legal system and security of property rights, bureaucratic quality, rule of law and corruption in government. Rule of law is an indicator of state legal capacity. Corruption in government and bureaucratic quality show the level of bureaucratic and administrative quality. Quality of legal system and security of property rights refer to the strength of legal system in transferring the property rights to the respective owner, without delaying the process and creating any further hurdles. It comprises of legal and political environment, physical property rights and intellectual property rights. All the indices are continuous variables and possess a specific data range which shows the lowest and highest level/strength of that specific index. A lower score in rule of law, bureaucratic quality and legal system and property rights index implies poor institutional quality, and higher score shows high institutional quality. However, corruption in government index follows the reverse pattern that its lower score corresponds to the high institutional quality, and higher score refers to the poor institutional quality. First, we have investigated whether the cross-sectional dispersion in the institutional quality indices increases or decreases overtime by testing the sigma convergence (see appendix, Table A1). Analysis of all the indices show that the cross-sectional dispersion in diminishing overtime, for the whole sample and period, indicating the strong evidence of sigma convergence in institutional quality measure. The results are statistically significant, but the speed of convergence varies across the type of institutional quality indicator. The coefficient of bureaucratic quality is high, followed by rule of law, legal system and property rights, and lastly corruption in government.

To observe the trends of all the indices we have used the mean and cross-sectional dispersion (coefficient of variation, CV) by

comparing economies at different stages of development over the sample time period (see Table A2 & A3). Decrease (increase) in cross-sectional dispersion (CV) of rule of law, bureaucratic quality and legal system and property rights indices will lead to the improvement (deterioration) in institutional quality measure and implies institutional quality convergence (non-convergence). Reverse is true for corruption in government. Institutional quality convergence indicates that countries with poor quality institutions tend to catch-up the countries having better institutional quality. Table A2 and A3 provided in the appendix show that the gap between advanced and poor economies decreases overtime implying institutional quality convergence. First, quality of legal system and security of property rights index has experienced improvements for the whole sample and rest of the economies, indicating the strong institutional quality convergence. The index shows decrease in the cross-sectional dispersion, approximately in all the economies, is generally monotonic until 2000, picked up again or becomes stable, indicating that either the institutional quality convergence is stopped or decelerated. Secondly, bureaucratic quality index shows that cross-sectional dispersion has decreased in whole sample and rest of the economies (except transition economies) again indicating the convergence of institutional quality. The decrease in cross-sectional dispersion is monotonic until 1995, approximately in all the economies (except developing economies). Thirdly, rule of law index has experienced improvements for the whole sample and rest of the economies exhibiting the strong institutional quality convergence. The cross-sectional dispersion has decreased in all the country groups, (except G7) and is monotonic until 1995. Finally, corruption in government index has also revealed the improvement in cross-sectional dispersion for the whole sample and rest of the world showing the strong institutional quality convergence. However, the cross-sectional dispersion is monotonic until 1990 in most of the economies and has also experienced the deterioration in transition economies which shows that the index has not improved in transition economies. It is evident that all the indices on aggregate show improvement, except for transition economies which show greater variability in institutional quality at the end of the period. And the cross-sectional dispersion on aggregate is generally monotonic until 1995, which follows the period of end of Cold War. A significant acceleration in the speed of institutional quality convergence is found from the end of cold war in early 1990s. However, the effect is weakened for the transition and developing economies in the new millennium.

Convergence of real per capita GDP

Cross-sectional dispersion of real per capita GDP among the selected country groups in a descriptive style are provided in the appendix (Table A4). Results exhibit significant variations across each group of countries due to their diverse structural characteristics, e.g. cultural, social, political, economic and geographical attributes. The analysis found evidence of sigma convergence in the whole sample, but the dispersion is not decreased significantly. All the developed economies (advanced, G7, European Union) show no strong evidence of sigma convergence, which strictly follows Mulligan and Sala-i-Martin (1993) that "beta convergence is necessary, but not sufficient condition for sigma convergence", because economies after importing technology from the developed nations boost-up their productivity by efficiently utilizing their unused resources to a level higher than the developed economies. The logic behind the scenario is that if better rule of law, political stability, voice and accountability prevails in a country, then importing technology from abroad and efficient use of human capital lead to the higher growth rates of the nations. However, our results favor the scenario that the developing and transition economies have gained remarkable output gains by improving the quality of institutions which implies

Table 1
2SLS estimates of institutional quality convergence.

	Fraser Institute Measures Legal System & Property Right	Bureaucratic Quality	ICRG Measures Rule of Law	Corruption in Government
Panel (a): Unconditional Convergence				
Initial Value	-0.002***	-0.004***	-0.003***	-0.003***
	(0.001)	(0.001)	(0.001)	(0.001)
Constant	0.011***	0.025***	0.022***	0.024***
	(0.003)	(0.004)	(0.002)	(0.004)
F Statistics	14.134***	32.365***	54.825***	12.764***
R. Squared	0.125	0.212	0.234	0.123
Obs.	90	104	104	104
RMSE	0.022	0.042	0.03	0.034
1st stage F stats	86.02***	1454.378***	1657.434***	2234.659***
Panel (b): Conditional Convergence				
Initial Value	-0.006***	-0.008***	-0.007***	-0.005***
	(0.001)	(0.001)	(0.001)	(0.001)
Constant	0.024*	0.045	0.034*	0.023
	(0.004)	(0.013)	(0.012)	(0.015)
F Statistics	8.215***	9.023***	12.934***	5.75***
R. Squared	0.445	0.605	0.756	0.434
Obs.	77	84	84	84
RMSE	0.008	0.017	0.012	0.013
1st stage F stats	62.214***	301.194***	456.76***	390.489***

Notes: *the dependent variable is the average annual growth rate of institutional quality measure. Symbols *, ** and *** stand for significance level at 10, 5 and 1 percent, respectively. Standard errors are in the parentheses. The control variables including the instruments used in conditional convergence regression include initial values of institutional quality measure, real per capita GDP growth rate, polity2 (political democracy), human capital, investment, R&D, governance index, religion in politics, ethnic fractionalization, geography (log of distance from the equator in KM), legal origin dummies.*
Source: Authors' estimates.

strong evidence of sigma convergence. Finally, the cross-sectional dispersion decreases in all developed economies until 2005 but developing follows the same pattern until 2010. Interestingly, the major advanced economies (G7) show weak evidence of sigma convergence for the sample period.

5.2. Empirical analysis

Our study is concerned with the evolution and change of institutions across countries and over time. We shall check whether contemporary differences in institutional quality levels between countries have become wider or narrower? The results of two-stage least square (2SLS) due to endogeneity between growth and institutional quality reveal strong evidence of convergence among countries of our sample (see Table 1). Notably, the speed of convergence is higher in conditional convergence estimates.

Results of growth convergence based on panel regressions are presented in Table 2 which are comprised of four panels, namely, sigma, unconditional beta, conditional beta and conditional beta (when only significant independent variables are included in the regression). We estimated simple OLS model 1, 2 and 3 for sigma, unconditional beta and conditional beta convergence, respectively. The panel of six selected group of countries is used in the estimation, namely, Advanced, Developing, East Asian Tigers, European Union, G7 and Transition Economies. *Panel (a)* indicates results of sigma convergence (non-convergence) that whether the cross-sectional dispersion of real per capita GDP is decreased (increased) overtime. The result shows that there exists significant negative trend among all economies (except G7 and European Union), which shows strong evidence of sigma convergence. G7 and European Union economies show negative but insignificant trend indicating non-convergence. *Panel (b)* shows unconditional beta convergence that how the poor economies of the world are catching-up the advanced countries in terms of real per capita GDP. The results suggest significant negative coefficients for all the economies, which imply strong evidence of unconditional beta convergence. But developing and transition economies reveal smaller coefficients which indicates slow speed of convergence. European Union economies show significant coefficient showing the strong evidence of unconditional beta convergence. Therefore, their results also coincide with the theory of Mulligan and Sala-i-Martin (1993) that is "beta convergence is necessary, but not sufficient condition for sigma convergence". There exists beta convergence, but no evidence of sigma convergence is found for the European Union region. Normally the result of unconditional convergence tends to be slower due to the diverse structural characteristics of countries e.g., law and order, culture, social, economic, and political, religious, and ethnic conflicts. Therefore, economists and policy makers suggest that by imposing conditions of homogenous structural characteristics, the speed of convergence can be enhanced, implies convergence in the long run. Panel (c) show result of conditional beta convergence, which indicates the significant negative coefficient for all the economies implying strong evidence of conditional beta convergence. The speed of convergence in both conditional and unconditional

Table 2
Panel data estimates of real per capita GDP convergence.

Advanced		Developing	East Asian	European Union	G7	Transition
Panel (a): Sigma convergence						
Trend	-0.0223	-0.0475	-0.0548	0.01	-0.0108	-0.1765
S.E.	0.0102	0.0198	0.0249	0.0163	0.0101	0.0798
Constant	2.7435	5.1144	2.8758	2.4792	1.3951	9.1166
P-value	0.0357**	0.0229**	0.0354**	0.5422	0.293	0.0348**
R Squared	0.1389	0.1608	0.1392	0.0125	0.0368	0.1402
Panel (b): Unconditional convergence						
Initial value	-0.9044	-0.825	-0.9198	-1.0096	-0.9784	-0.8861
S.E.	0.0273	0.0322	0.4376	0.0278	0.0819	0.0621
Constant	1.7945	1.7535	3.4945	2.3335	1.3849	2.0574
S.E.	0.1283	0.1535	3.5491	0.1308	0.3163	0.4708
P-value	0.0179**	0.0211**	0.0383**	0.0281**	0.0001*	0.0257**
R Squared	0.5367	0.4069	0.0463	0.6382	0.399	0.3117
Panel (c): Conditional convergence						
Initial value	-1.0245	-0.8951	-10.1608	-1.0155	-1.1104	-2.3439
S.E.	0.0407	0.1121	11.3262	0.0435	0.2791	0.5507
Constant	-9.4586	-5.1459	156.7321	-10.6536	3.4454	44.8969
S.E.	2.9034	4.0415	96.0064	3.8154	15.3586	50.3485
P-value	0.0018*	0.0357**	0.0223**	0.0179**	0.0018*	0.0892***
R Squared	0.6722	0.5004	0.5709	0.7548	0.6389	0.889
Panel (d): Conditional Sigma convergence						
Initial value	-1.0273	-0.9382	-1.8829	-0.9735	-1.1993	-0.919
S.E.	0.0373	0.0316	0.8523	0.0344	0.1134	0.0647
Constant	-6.8316	-2.997	30.0649	-11.8798	-11.9896	7.6049
S.E.	2.0767	0.6398	808.8772	2.2585	3.59	1.8814
P-value	0.0739***	0.0025*	0.0021*	0.0225**	0.0011*	0.0001*
R Squared	0.6727	0.5141	0.3073	0.7558	0.6226	0.7311

Notes: *the dependent variable is the annual growth rate of real per capita GDP. Symbols *, ** and *** stand for significant level at 1, 5 and 10 percent, respectively. Standard errors are in the parentheses. The control variables used in the conditional convergence regression include initial value of per capita GDP growth rate, human capital, saving, investment, trade openness, R&D expenditure, inflation rate, population growth rate, institutional quality, economic freedom, polity2, govt. stability, military in power and politics.*
Source: Authors' estimates.

beta measures is significantly increased by incorporating the role of control variables mainly in the developing, European and transition economies. The results are compatible with the theory of conditional convergence.

Results of institutional quality convergence based on OLS panel regressions estimates from the Eq. (4) and 5 are presented in Table 3 comprising two panels, namely, unconditional, and conditional beta convergence. Panel (a) represents unconditional beta convergence in institutional quality, that is how the poor-quality institutions are catching-up the high-quality institutions of advanced economies, in terms of rule of law, bureaucratic quality, legal system and property rights, economic and political structure, and other measures of political stability. Result shows significant negative value for all economies (except transition economies) indicating the strong evidence of unconditional convergence in institutional quality. However, the result of transition economies is negative but insignificant implying non-convergent behavior of institutional quality. The results of conditional beta convergence are presented in *panel (b)*, which show significant negative value of institutional quality measure revealing the strong evidence of convergence. Also, the speed of convergence is increased in the respective converging economies, which strongly correlates with the results of real per capita GDP and theory of conditional convergence.

We have reached to the conclusion that there exists both types of convergence in the selected countries and many results coincide with each other e.g., there is slow process of unconditional convergence in terms of both growth and institutional quality. The speed of convergence has increased for both growth and institutional quality by incorporating the role of control variables which indicates conditional convergence. For example, there exists sigma convergence, unconditional beta convergence and conditional beta convergence with high speed in both growth and institutional quality. Such empirical link between the convergence of growth and institutional quality indicates that institutions strongly play an empirical role in the growth process of the economy. However, when we observe the speed of growth and institutional quality convergence (conditional convergence results of growth and institutions) across country groups with different levels of development, an interesting link is unveiled which bears strong implications. East Asian and high-income countries exhibit high speed of growth and low speed of institutional quality convergence, and reverse is true for developing countries. These findings show that despite the institutional quality being a significant determinant of growth, its convergence cannot fully explain cross-country income differences. We shall now proceed to formally discuss whether the institutional quality convergence leads to growth convergence?

5.2.1. Identification of linkages between growth and institutional quality convergence

Finally, to address the third research question we have performed cross-sectional analysis based on the estimates presented in the Table 4, we classified our analysis into two panels, (a) growth convergence, and (b) institutional quality convergence. Panel (a) consists of four types of results, namely, sigma, unconditional beta, conditional beta, and conditional sigma. However, institutional quality convergence in panel (b) comprises of just unconditional and conditional beta convergence because we already have provided the sigma convergence analysis of all institutional quality measures (see Table 1). Results of sigma convergence in panel (a) show significant value of coefficient which implies strong evidence of convergence. Results of unconditional, conditional beta, and conditional beta (when only significant control variables are included) show significant value for all groups and estimates and indicate the strong evidence of respective growth convergence across the world. Notably the coefficient of conditional beta convergence is higher than unconditional, which indicates higher speed of convergence. *Panel (b)* shows two types of convergence results, unconditional and conditional convergence of institutional quality. The results of unconditional and conditional beta convergence provide significant coefficients confirming the strong evidence of institutional quality convergence. Hence the poor-quality institutions are catching-up the higher quality institutions of advanced economies. Again, note that as in case of growth convergence provided in panel (a), the coefficient of conditional institutional convergence is higher than unconditional which indicates higher speed of convergence. The

Table 3
Panel data estimates of institutional quality convergence.

	Advanced	Developing	European Union	Transition
Panel (a): Unconditional convergence				
Initial value	-0.6002	-1.5901	-0.5487	-0.1942
S.E.	(0.0812)	(0.2982)	(0.1093)	(0.3513)
Constant	3.1576	5.7264	2.7569	1.0821
S.E.	(0.422)	(0.8676)	(0.5281)	(1.2403)
P-value	0.0010*	0.0082*	0.0015*	0.5892
R Squared	0.6456	0.4867	0.5018	0.0214
Panel (b): Conditional convergence				
Initial value	-0.8293	-1.7729	-0.9621	-0.1359
S.E.	(0.2104)	(0.6581)	(0.1037)	(0.359)
Constant	2.9669	-4.9211	10.3697	1.061
S.E.	(2.8513)	(7.7074)	(4.2756)	(1.2476)
P-value	0.0333**	0.0489**	0.0052*	0.578
R Squared	0.8515	0.943	0.7946	0.0809

Notes: *the dependent variable is the average annual growth rate of institutional quality measure. Symbols * and ** stand for significant level at 1 and 5 percent, respectively. Standard errors are in the parentheses. The control variables used in conditional convergence regression include initial values of institutional quality, real per capita GDP growth rate, polity2 (political democracy), human capital, investment, R&D, governance index, religion in politics, ethnic fractionalization, geography (log of distance from the equator in KM), legal origin dummies.*
Source: Authors' estimates.

Table 4
Growth & institutional quality convergence.

Panel (a): Growth Convergence				
	Sigma	Unconditional	Conditional	Conditional sigma
Trend/Initial value	-0.5186	-1.7015	-2.0503	-0.5186
S.E.	(0.0202)	(0.1022)	(0.4747)	(0.3425)
Constant	-3.9541	0.7102	1.2739	-6.6462
S.E.	(0.3646)	(0.4055)	(6.5882)	(4.0468)
P-value	0.0136**	0.0010*	0.0001*	0.0000*
R Squared	0.0316	0.1227	0.7387	0.7264
Panel (b): Institutional Quality Convergence				
		Unconditional	**Conditional**	
Initial value		-0.6649	-1.1818	
S.E.		(0.0749)	(0.1288)	
Constant		3.2496	3.19	
S.E.		(0.3279)	(1.1591)	
P-value		0.0000*	0.0001*	
R Squared		0.4068	0.7609	

Notes: *the dependent variable is the growth rate of real per capita GDP. Symbols * and ** stand for significant level at 1 and 5 percent, respectively. Standard errors are in the parentheses. The control variables including the instruments used in the conditional growth convergence regression contain initial value of per capita GDP growth rate, human capital, saving, investment, trade openness, R&D expenditure, population growth rate, institutional quality, economic freedom, polity2, govt. stability, military in power and politics. Variables used in the conditional institutional quality convergence regression contain initial values of institutional quality measure, real per capita GDP growth rate, polity2 (political democracy), human capital, investment, R&D, governance index, religion in politics, ethnic fractionalization, geography (log of distance from the equator in KM), legal origin dummies.*
Source: Authors' estimates.

findings of unconditional and conditional beta convergence exhibit an empirical link which validates the existing literature in terms of vitality of institutional quality for economic growth and convergence. Looking at the speed of growth and institutional quality convergence in Table 2 and 3 (only conditional convergence coefficients for better focus) it is evident that dynamics of convergence speed in the growth and institutional quality are clearly different at different levels of development stages. Based on these findings we empirically show that the institutional quality convergence alone cannot ensure the growth convergence. These findings unveil the potential lock-ins and path dependence of growth trajectories of poorer countries. Probably, there must be complementary economic inertia in countries in order to catch-up rich economies.

Conclusion and policy implications

The empirical analysis of this study is based on 120 countries categorized into six groups in terms of their development levels. Findings confirm the heterogeneities of countries in terms of speed of convergence. The growth convergence results indicate strong evidence of convergence with varying speed of convergence across development stages of countries. In the unconditional growth convergence, as compared to the developed countries, developing and transition economies show relatively slow speed of convergence. Speed of growth convergence is found to be significantly higher in the conditional convergence mainly in case of East Asian and transition economies, while the developing countries show lowest speed of growth convergence in the conditional convergence estimates. These findings are aligned with the observed south-south divergence as discussed by Frankema (2025). Moreover, these results provide practical policy implications for rising disparities among global south countries.

Empirical results of the institutional quality convergence reveal striking trends. The gap between poor and advanced economies is diminishing over time except the transition economies, however the speed of institutional convergence shows contradictory trajectories as compared to the speed of growth convergence. Speed of institutional convergence is highest in developing countries followed by European Union, advanced, and transition economies, respectively. The empirical dynamics of growth and institutional quality convergence show that both the growth and institutional quality converge faster in conditional convergence model indicating that growth and institutions converge in countries with similar characteristics. However, this co-movement does not necessarily imply that this phenomenon holds for all development stages. These findings unveil a decoupling between speed of growth and institutional quality convergence and potentially explain another aspect of south-south and north-south divergence in terms of rising disparities in institutional quality. These empirical links indicate that despite the institutions being a fundamental driving force of growth their convergence cannot fully explain cross-country income differences. Such decoupling link between speed of growth and institutional convergence indicate the path dependence and lock-ins of growth trajectories in developing countries.

Based on the empirical findings of the study, country- and region-specific policies are recommended in order to address the disparities and catch-up the high-income countries. East Asian and Transition economies are converging in terms of growth, but the speed of institutional convergence is insignificant. It is suggested that these countries should follow concrete national and regional level policies in terms of innovation and regional linkages in order to avoid future disparities. Despite showing highest speed of institutional convergence, the developing countries show lowest speed of growth convergence. These findings imply that despite the institutional quality being a fundamental cause of growth, it alone cannot improve the income levels of these countries. The developing countries should devise effective policies in order to ensure technology transfer and local capabilities to upgrade their economies through

infrastructure development, access to finance and policy consistencies. European Union countries show moderate speed of convergence both in growth and institutional quality. Further encouraging the regional integration in terms of innovation and economic linkages could benefits these countries to catch-up both in growth and institutional quality. Broadly, the empirical findings highlight the rising disparities in institutions and growth hence calls for effective regional and global policy frameworks in order to tackle the global income inequalities.

CRediT authorship contribution statement

Muhammad Safdar: Formal analysis, Data curation, Conceptualization, Investigation, Writing – original draft, Methodology. **Ahmad Nawaz:** Supervision, Writing – review & editing, Resources, Validation, Software, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix

Table A1
Sigma convergence of institutional quality.

(A) Quality of legal system & security of property rights index				
Dependent Variable: SIGMA				
Method: Least Squares				
Variable	Coefficient	S.E	t-value	Prob.
C	2.7228	0.2502	10.8844	0.0000
Trend	-0.3826	0.0407	-9.4026	0.0000
R-squared	0.4368			
Prob. (F-statistic)	0.0000			
(B) Bureaucratic Quality Index				
Dependent Variable: SIGMA				
Method: Least Squares				
Variable	Coefficient	S.E	t-value	Prob.
C	4.6986	0.6901	6.8082	0.0000
Trend	-1.4097	0.2389	-5.9003	0.0000
R-squared	0.2421			
Prob. (F-statistic)	0.0000			
(C) Rule of Law Index				
Dependent Variable: SIGMA				
Method: Least Squares				
Variable	Coefficient	S.E	t-value	Prob.
C	4.4980	0.3136	14.3417	0.0000
Trend	-0.9017	0.0717	-12.5808	0.0000
R-squared	0.5900			
Prob. (F-statistic)	0.0000			
(D) Corruption in Government Index				
Dependent Variable: SIGMA				
Method: Least Squares				
Variable	Coefficient	S.E	t-value	Prob.
C	1.2078	0.2854	4.2320	0.0000
Trend	-0.3930	0.0671	-5.8540	0.0000
R-squared	0.2375			
Prob. (F-statistic)	0.0000			

Source: Authors' estimates.

Table A2
Dynamics of institutional quality around the world 1990–2015.

Year		1990	1995	2000	2005	2010	2015
Panel (a): Quality of Legal System and Security of Property Rights Index							
Whole Sample	Mean	6.42	6.95	7.16	6.54	6.47	6.42
	CV	0.31	0.43	0.26	0.26	0.23	0.23
Advanced Economies	Mean	7.52	8.01	8.26	7.71	7.46	7.41
	CV	0.16	0.14	0.14	0.12	0.13	0.13
Developing Economies	Mean	4.19	5.2	5.24	4.99	5.23	5.15
	CV	0.37	0.2	0.29	0.28	0.26	0.27

(*continued on next page*)

Table A2 (*continued*)

Year		1990	1995	2000	2005	2010	2015
East Asian Tigers	Mean	6.21	7.37	7.24	7.86	7.69	7.64
	CV	0.13	0.21	0.18	0.11	0.14	0.14
European Union	Mean	7.49	7.54	7.77	7.19	7.02	6.98
	CV	0.14	0.17	0.17	0.16	0.15	0.15
G7 Economies	Mean	8	8.25	8.69	7.71	7.45	7.37
	CV	0.05	0.12	0.08	0.11	0.1	0.11
Transition Economies	Mean	5.21	4.57	5.37	5	5.32	5.39
	CV	0.36	0.23	0.28	0.14	0.11	0.1
Panel (b): Bureaucratic Quality Index							
Whole Sample	Mean	2.72	2.99	2.92	2.84	2.85	2.84
	CV	0.45	0.37	0.36	0.36	0.37	0.36
Advanced Economies	Mean	3.49	3.69	3.67	3.59	3.59	3.59
	CV	0.21	0.13	3.13	0.16	0.16	0.16
Developing Economies	Mean	1.66	1.91	2.01	1.96	1.98	1.92
	CV	0.62	0.5	0.34	0.34	0.34	0.38
East Asian Tigers	Mean	2.83	3.5	3.33	3.33	3.33	3.33
	CV	0.23	0.21	0.17	0.18	0.18	0.18
European Union	Mean	3.15	3.4	3.31	3.22	3.22	3.22
	CV	0.31	0.24	0.25	0.25	0.25	0.25
G7 Economies	Mean	3.83	3.93	3.86	3.64	3.64	3.64
	CV	0.11	0.05	0.1	0.18	0.16	0.1
Transition Economies	Mean	1.88	1.88	1.64	1.59	1.68	1.71
	CV	0.45	0.45	0.41	0.42	0.45	0.17

Source: Authors' calculations.

Table A3
Dynamics of institutional quality around the world 1990–2015.

Year		1990	1995	2000	2005	2010	2015
Panel (c): Rule of Law							
Whole Sample	Mean	3.9	5.19	4.64	4.49	4.43	4.39
	CV	0.45	0.23	0.28	0.26	0.26	0.27
Advanced economies	Mean	4.93	5.88	5.37	5.21	5.19	5.18
	CV	0.3	0.06	0.15	0.13	0.12	0.13
Developing Economies	Mean	2.31	3.94	3.64	3.4	3.31	3.23
	CV	0.51	0.3	0.36	0.34	0.34	0.34
East Asian Tigers	Mean	3.67	5.67	5	4.88	5	5
	CV	0.34	0.11	0.2	0.04	0.04	0.03
European Union	Mean	4.85	5.83	5.04	4.95	4.91	4.89
	CV	0.25	0.07	0.18	0.16	0.18	0.18
G7 Economies	Mean	5.33	5.92	5.68	5.07	5	4.96
	CV	0.1	0.04	0.08	0.12	0.1	0.1
Transition Economies	Mean	2.9	2.9	3.64	3.85	3.75	3.64
	CV	0.37	0.37	0.31	0.22	0.17	0.17
Panel (d): Corruption in Government							
Whole Sample	Mean	4.01	4.2	3.52	3.16	3.3	3.3
	CV	0.37	0.32	0.38	0.41	0.39	0.38
Advanced economies	Mean	4.97	5	4.33	4.08	4.15	4.16
	CV	0.22	0.19	0.25	0.25	0.24	0.24
Developing Economies	Mean	2.66	2.92	2.54	2.08	2.32	2.28
	CV	0.41	0.32	0.34	0.33	0.27	0.26
East Asian Tigers	Mean	3.67	4.67	3.42	3.67	3.96	3.98
	CV	0.35	0.22	0.16	0.29	0.22	0.22
European Union	Mean	4.76	4.78	4.16	3.61	3.66	3.67
	CV	0.22	0.2	0.27	0.31	0.34	0.33
G7 Economies	Mean	5	4.86	4	4.08	4.22	4.22
	CV	0.13	0.23	0.29	0.24	0.2	0.21
Transition Economies	Mean	2.74	2.12	1.79	1.98	2.07	2.16
	CV	0.27	0.32	0.23	0.23	0.35	0.47

Source: Authors' calculations.

Table A4
Dynamics of real per capita GDP growth rate around the world 1985–2015.

Year		1990	1995	2000	2005	2010	2015
Whole Sample	Mean	1.53	1.95	2	3.91	3.91	2.81
	CV	2.49	2.66	2.57	0.73	0.98	1.26
Advanced Economies	Mean	2.66	2.91	3	4.34	2.85	2.08
	CV	0.62	0.87	0.74	0.47	0.75	1.59

(*continued on next page*)

Table A4 (*continued*)

Year		1990	1995	2000	2005	2010	2015
Developing Economies	Mean	-0.3	1.23	1.48	1.56	3.36	3.63
	CV	-18.38	3.44	3.11	2.04	1.08	1.24
East Asian Tigers	Mean	1.76	5.86	4.02	7.23	5.21	8.41
	CV	2.3	0.41	0.94	0.09	0.32	0.49
European Union	Mean	2.49	2.36	3.62	4.52	3.67	1.52
	CV	0.35	1.42	0.61	0.41	0.86	1.53
G7 Economies	Mean	3.29	1.93	1.89	3.2	1.46	2.31
	CV	0.44	1.19	0.28	0.22	0.56	0.67
Transition Economies	Mean	-0.82	-0.79	-1.75	5.97	7.86	3.86
	CV	-6.26	-17.12	-6.18	0.45	0.72	0.77

Source: Authors' calculations.

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